Midterm 1 Solutions

ORF 309

Name: ______________________________

Instructions: This midterm has 5 questions for a total of 100 points. Partial credit will be awarded for partial proofs or arguments. You may freely use basic facts from class, such as laws of probability, properties of conditional expectation, or manipulations with pmfs/pdfs. However, students must provide adequate justifications for especially important steps. Put another way, try your best to demonstrate awareness that you are applying the correct facts in the correct way by explicitly mentioning the results you are applying, indicating the objects they are applied to, and checking the required assumptions.

Good Luck!

Question	Points	Score
1	25	
2	20	
3	5	
4	30	
5	20	
Total:	100	

1. (Drawing Cards) Suppose you successfully draw cards independently and uniformly at random from a standard deck of 52 cards, i.e., 4 suits (hearts, diamonds, spades, and clubs) each with 13 denominations (labelled $A, 2, \ldots, 10, J, Q, K$).

   (a) (5 points) How many draws do you expect to perform before arriving at a heart if successive draws are done *with replacement*?
   Solution: This problem was solved in class. The number of draws X is Geometric(p) random variable with $p = 1/4$. Hence, from our work in class, we know $\mathbb{E}[X] = 1/p = 4$.

   (b) (10 points) How many draws do you expect to perform before arriving at a heart if successive draws are done *without replacement*?
   Solution: This problem was solved in class. For $n = 0, \ldots, 39$, let X_n be the number of draws BEFORE getting a heart if the deck currently still has all 13 hearts but n non-hearts have been removed. In particular, we have $X_{39} \equiv 0$, and letting Y indicate whether the next event is a heart, we have by law of total expectation

$$\begin{aligned}\mathbb{E}[X_n] &= \mathbb{E}[X_n|Y=1]\mathbb{P}(Y=1) + \mathbb{E}[X_n|Y=0]\mathbb{P}(Y=0)\\ &= 0 \cdot \mathbb{P}(Y=1) + [1 + \mathbb{E}[X_{n+1}]]\frac{39-n}{52-n}\\ &= [1 + \mathbb{E}[X_{n+1}]]\frac{39-n}{52-n}.\end{aligned}$$

   We saw that $\mathbb{E}[X_n] = \frac{39-n}{14}$ satisfies this recurrence with zero boundary condition at $n = 39$, so the desired answer is

$$1 + \mathbb{E}[X_0] = 1 + \frac{39}{14} = \frac{53}{14} \approx 3.786.$$

   (c) (10 points) What is the probability of getting an ace (A) before a face (J, Q, K) if successive draws are done *with replacement*? How about *without replacement*? (Hint: Intuitive reasoning for the answers here is sufficient.)
   Intuitive (Acceptable) Solution: All that matters is the moment you "hit the boundary," i.e., the moment you draw a card that is an ace or a face. No matter what occurred before, whether with or without replacement, it is intuitively clear that reduction of sample space to the boundary will be uniform over all 16 aces and faces, so the probability of drawing one of the 4 aces will be $4/16 = 1/4$.
   Brute force solution (sketch): Let E be the event of drawing an ace before a face, and let A, F, N be the events of drawing an ace, face, or neither on the next card.
   If with replacement, then law of total probability gives

$$\begin{aligned}\mathbb{P}(E) &= \mathbb{P}(E|A)\mathbb{P}(A) + \mathbb{P}(E|F)\mathbb{P}(F) + \mathbb{P}(E|N)\mathbb{P}(N)\\ &= 1 \cdot \frac{4}{52} + 0 \cdot \frac{12}{52} + \mathbb{P}(E) \cdot \frac{36}{52}.\end{aligned}$$

Solving for $\mathbb{P}(E)$ gives $\mathbb{P}(E) = 1/4$.

If without replacement, we can do a probability version of the argument above for hearts, letting A_n be the event the nth draw is an ace and N_n be the event of neither on the first $n-1$ draws. Then the desired probability is

$$\mathbb{P}\left(\cup_{k=1}^{37} N_n \cap A_n\right) = \sum_{k=1}^{37} \mathbb{P}\left(N_n \cap A_n\right) = \sum_{k=1}^{37} \mathbb{P}\left(A_n | N_n\right) \mathbb{P}(N_n)$$
$$= \sum_{k=1}^{37} \frac{4}{52-(n-1)} \cdot \frac{\binom{36}{n-1}}{\binom{52}{n-1}} = \frac{1}{4},$$

where the first equality follows because the events $N_n \cap A_n$ are disjoint and the second equality follows by law of multiplication.

2. (Trick Coins) Determine a blank coin by painting each side heads or tails with probability 1/2 independently for each side.

   (a) (10 points) Given you flip heads with the painted coin, what is the probability your next flip is also heads?

   Solution: This problem was solved in class. You do not need to use Bayes' theorem; indeed, we are not asked to update old likelihoods with new information, we are simply computing a conditional expectation with various cases. Let H be the event of flipping one heads and HH the event of flipping two heads. Using $HH \subset H$, we have

$$\mathbb{P}(HH|H) = \frac{\mathbb{P}(HH, H)}{\mathbb{P}(H)} = \frac{\mathbb{P}(HH)}{\mathbb{P}(H)} = \frac{0 \cdot 1/4 + 1 \cdot 1/4 + (1/2)^2 \cdot 1/2}{0 \cdot 1/4 + 1 \cdot 1/4 + 1/2 \cdot 1/2} = \frac{2+1}{2+2} = \frac{3}{4},$$

   where the main middle equality is law of total probability in both the numerator and denominator based on the partition of types of painted coins (0 heads, 1 heads, and 2 heads).

   Suppose the painting procedure above produced a double-headed coin, which gets comingled with 999 other fair coins. Suppose you pick a coin uniformly at random from the 1000 total and flip 10 heads in a row with the chosen coin.

   (a) (10 points) What is the probability you chose the painted double-headed coin?

   Solution: Let A be the event that the painted double-headed coin is picked, and let B denote the event that you flip 10 heads in a row with the chosen coin. We are asked to find the conditional probability $\mathbb{P}[A|B]$. By assumptions,

$$\mathbb{P}[A] = \frac{1}{1000} = 1 - \mathbb{P}[A^c],$$

   and it is easy to calculate that

$$\mathbb{P}[B|A] = 1, \quad \mathbb{P}[B|A^c] = \left(\frac{1}{2}\right)^{10} = \frac{1}{1024}.$$

   Using Bayes' theorem, we get

$$\mathbb{P}[A|B] = \frac{\mathbb{P}[B|A]\mathbb{P}[A]}{\mathbb{P}[B|A]\mathbb{P}[A] + \mathbb{P}[B|A^c]\mathbb{P}[A^c]} = \frac{1 \times \frac{1}{1000}}{1 \times \frac{1}{1000} + \frac{1}{1024} \times \frac{999}{1000}} = \frac{1024}{2023} \approx 0.50617894216.$$

3. (Lottery) You enter a game that requires you to draw two ping-pong balls from a bag containing four that are labelled $1, 2, 3, 4$. If at least one of the two you draw is strictly greater than 2, you win \$1; otherwise, you lose \$1. Let X be your net profit and let Y be the number on the first ball drawn.

   (a) (5 points) Compute $\mathbb{E}[X|Y]$ (intuitive reasoning for the expression is sufficient).

   Solution: This is from Homework 3. We have

$$\mathbb{E}[X|Y] = 1_{\{3,4\}}(Y) + \frac{1}{3}1_{\{1,2\}}(Y).$$

   To see why, if you first draw a 3 or 4, then you are guaranteed to win \$1 (the first term). However, if you first draw a 1 or 2, then we may reason based on reduction of sample space as follows: 1/3 of the time you draw the remaining small ball and lose money, while 2/3 of the time you draw a bigger ball and win, so the expected gain here is $2/3 \cdot 1 + 1/3 \cdot (-1) = 1/3$ (the second term).

4. (Poisson processes)

Let $(N_t)_{t\geq 0}$ be a Poisson process of rate $\lambda > 0$, and let $\tau \sim \text{Exp}(\rho)$ be an independent exponential random variable of parameter ρ.

(a) (10 points) Suppose $X \sim \text{Exp}(\mu)$ where $\mu := \lambda + \rho$. Compute $\mathbb{E}[X^k]$ for $k \in \mathbb{N}$.
Solution: This problem was solved in class. A brute force solution is to integrate by parts repeatedly. However, we do the "parameter trick": note first that by definition of a probability density function, we have

$$\int_0^\infty e^{-\mu x}\, dx = \frac{1}{\mu}.$$

Now differentiating both sides with respect to μ k-times gives

$$\int_0^\infty (-1)^k x^k e^{-\mu x}\, dx = (-1)^k \frac{k!}{\mu^{k+1}}$$

Cancelling the $(-1)^k$ from both sides and bringing one copy of the μ to the left side gives

$$\mathbb{E}[X^k] = \int_0^\infty x^k\, \mu e^{-\mu x}\, dx = \frac{k!}{\mu^k}$$

(b) (10 points) Compute $\mathbb{P}(N_\tau = k)$ for all $k \geq 0$.
Solution 1: We can brute force compute

$$\mathbb{P}(N_\tau = k) = \mathbb{E}[\mathbb{P}(N_\tau = k|\tau)] \overset{\text{independence}}{=} \mathbb{E}\left[e^{-\lambda\tau}\frac{(\lambda\tau)^k}{k!}\right] = \int_0^\infty \frac{(\lambda t)^k}{k!}\rho e^{-(\rho+\lambda)t} dt$$
$$= \frac{\lambda^k}{k!}\cdot\frac{\rho}{\rho+\lambda}\cdot\int_0^\infty t^k\,(\rho+\lambda)e^{-(\rho+\lambda)t}\, dt \overset{\text{Part (a)}}{=} \left(\frac{\lambda}{\rho+\lambda}\right)^k\cdot\frac{\rho}{\rho+\lambda}.$$

Suppose the probability of observing at least one car during any 20 minutes is $\frac{609}{625}$. Assume arrivals occur with equal probability and independently "at every time."

(a) (10 points) What is the probability of seeing at least one car during any 5 minute period?

Solution: If $(N_t)_{t\geq 0}$ is the Poisson process counting cars observed with unknown rate $\lambda > 0$ cars per minute, then we are given that

$$\frac{609}{625} = \mathbb{P}(N_{20} \geq 1) = 1 - \mathbb{P}(N_{20} = 0) = 1 - e^{-\lambda\cdot 20}$$

One can then solve for λ and find the desired answer of $\mathbb{P}(N_5 \geq 1) = 3/5$.

For a shortcut, we can write $p := \mathbb{P}(N_5 \geq 1)$, and observe by independent increments that we have $(1-p)^4 = \mathbb{P}(N_{20} = 0) = 1 - \frac{609}{625} = \frac{16}{625}$, giving $p = 3/5$.

5. (Rock, paper, scissors). This classic two-player game is played as follows. There are three different hand gestures representing rock, paper, or scissors. In every round of the game, each player simultaneously makes one of the gestures (so neither player knows what the other is going to choose).

   If both players make the same gesture, the round is a tie. Otherwise, one of the players wins according to the following rules: **r** wins against **s**, **p** wins against **r**, and **s** wins against **p**. When the players tie, they keep playing additional rounds until one of them wins, at which point the game ends.

   As each player cannot know what the other will play, the best they can do is the following strategy: in each round, each player independently chooses one of the gestures **r, p, s** with equal probability (that is, each player and each round is independent).

   (a) (5 points) What is the probability that the first round is a tie?

   Solution $\frac{1}{3}$

   (b) (5 points) What is the expected number N of rounds played until someone wins?

   Solution N is geometric with success probability $\frac{2}{3}$, so $\mathbb{E}[N] = \frac{3}{2}$.

   (c) (5 points) What is the probability that the first player eventually wins the game?

   Solution $\sum_{k=1}^{\infty} \mathbb{P}(\text{Player 1 wins round k}) = \sum_{k=1}^{\infty} \left(\frac{1}{3}\right)^k = \frac{1}{2}$

   (d) (5 points) What is the expected number of times the first player plays **r** throughout the game?

   Solution If $X_k \in \{\mathbf{r}, \mathbf{p}, \mathbf{s}\}$ is the play by player 1 in round $k \geq 1$, then the desired quantity $\mathbb{E}[R]$ where $R = \sum_{k=1}^{N} 1_{\{\mathbf{r}\}}(X_k)$, where N is from part (b). Hence we compute

$$\mathbb{E}[R] = \mathbb{E}[\mathbb{E}[R|N]] \overset{\text{independence}}{=} \mathbb{E}[N \cdot \frac{1}{3}] = \frac{1}{3} \cdot \frac{3}{2} = \frac{1}{2}.$$